Code No: 136FB **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February/March - 2022
FABRICATION PROCESSES
(Common to CE, EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Sketch and explain the working of a crucible furnace. Give its applications.
b) What are the elements of Gating System? Explain with sketches. [8+7]

2.a) Explain various stages in investment casting process using suitable sketches. Also, list the applications of the process.
b) What are the casting defects? Explain the reasons for such defects and suggest suitable remedies. [8+7]

3.a) Explain the working principle of (i) spot welding (ii) flash butt welding in detail.
b) Briefly describe the Oxy-Acetylene welding technique with a neat sketch. [7+8]

4.a) Give the advantages and limitations of TIG welding over MIG welding.
b) Discuss various destructive and non-destructive methods of testing weld quality. [7+8]

5.a) Distinguish between punching and blanking operations. Also, design punch and dies used in these operations.
b) What are the specific merits of cold working over hot working? [8+7]

6.a) The shear strength of cold rolled steel is 300 MPa. Determine the punch force required for making a blank of 120 mm diameter from a strip of 3 mm thickness of the above material. What is the theoretical and maximum punch force required?
b) Sketch and explain wire drawing and Tube drawing processes in detail. Give their applications and advantages. [6+9]

7.a) Distinguish between cold and hot extrusion processes.
b) List out the advantages, disadvantages and limitations of extrusion. [8+7]

8.a) Discuss the effect of friction in open die and closed die forging processes.
b) What is meant by Upset Forging? Sketch and explain it in detail. Mention Forging defects and remedies. [8+7]

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